BTC DCA + Options Hedging

Multi-Timeframe Comparison

Generated 2026-02-19

Master Comparison

Strategy	Window	Return%	Ann.Ret%	Sharpe	MaxDD%	Rolls
BTC DCA	6mo	-3.4	-6.6	-2.028	48.4	-
SPY Buy&Hold	6mo	+7.5	+15.7	0.904	5.1	-
QQQ Buy&Hold	6mo	+6.4	+13.2	0.555	7.9	-
Cash Reserve	6mo	-41.0	-65.2	-2.048	49.0	-
Trailing Stop	6mo	-25.5	-44.5	-1.602	34.5	-
Adaptive Sizing	6mo	-40.3	-64.3	-2.033	48.4	-
BTC DCA+SPY PUT	6mo	-38.5	-62.1	-1.857	48.0	-
BTC DCA+IWM PUT	6mo	-38.3	-62.0	-1.850	47.6	-
BTC DCA+SPY COLLAR	6mo	-36.5	-59.7	-1.786	46.6	-
SPY PUT	6mo	+4.4	+9.0	0.390	5.1	6
IWM PUT	6mo	+13.4	+28.7	1.230	8.8	6
SPY COLLAR	6mo	+9.3	+19.4	1.514	3.9	6
BTC DCA	3mo	-4.4	-16.5	-2.124	34.0	-
SPY Buy&Hold	3mo	+4.0	+16.9	1.023	2.6	-
QQQ Buy&Hold	3mo	+0.3	+1.4	-0.143	5.7	-
Cash Reserve	3mo	-27.3	-72.0	-2.139	34.1	-
Trailing Stop	3mo	-21.3	-61.7	-2.817	24.1	-
Adaptive Sizing	3mo	-27.1	-71.7	-2.126	34.0	-
BTC DCA+SPY PUT	3mo	-25.9	-69.8	-1.900	34.6	-
BTC DCA+IWM PUT	3mo	-25.0	-68.4	-1.875	33.9	-
BTC DCA+SPY COLLAR	3mo	-24.7	-67.8	-1.836	34.0	-
SPY PUT	3mo	+2.4	+10.1	0.501	2.9	3
IWM PUT	3mo	+11.8	+56.2	2.463	4.8	3
SPY COLLAR	3mo	+4.8	+20.7	1.762	2.1	3
BTC DCA	1mo	-5.1	-46.7	-4.795	29.2	-

SPY Buy&Hold	1mo	+1.3	+16.6	0.958	2.6	-
QQQ Buy&Hold	1mo	-0.4	-4.4	-0.467	5.7	-
Cash Reserve	1mo	-25.9	-97.3	-4.825	29.3	-
Trailing Stop	1mo	-19.4	-92.5	-4.637	21.8	-
Adaptive Sizing	1mo	-25.7	-97.2	-4.797	29.1	-
BTC DCA+SPY PUT	1mo	-25.3	-97.0	-4.342	29.8	-
BTC DCA+IWM PUT	1mo	-25.8	-97.2	-4.540	30.0	-
BTC DCA+SPY COLLAR	1mo	-24.9	-96.8	-4.339	29.4	-
SPY PUT	1mo	+0.8	+9.8	0.463	2.6	1
IWM PUT	1mo	+0.0	+0.4	-0.149	4.8	1
SPY COLLAR	1mo	+1.6	+20.4	1.626	1.6	1

2. BTC DCA Results

Pairwise DCA strategy applied to BTC (Grayscale Bitcoin Mini Trust ETF) across three time windows. The strategy places paired sell-stop / buy-limit orders to reduce cost basis through mean-reversion captures.

BTC DCA -- 6mo Window

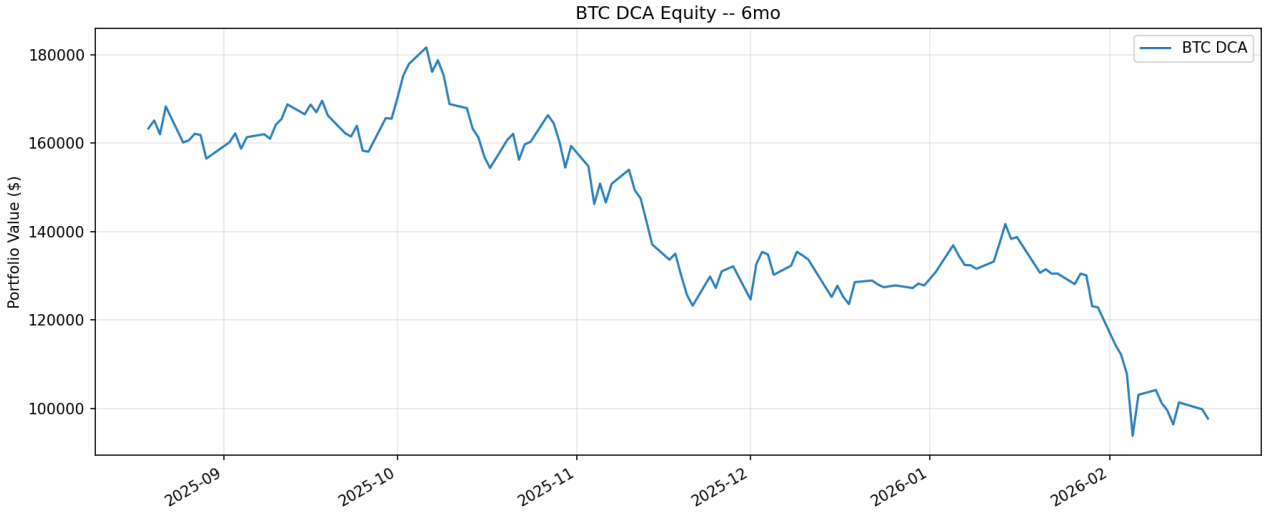
Return: -3.38% (Annualized: -6.64%)

Sharpe Ratio: -2.028

Max Drawdown: 48.37%

Pairs: 112 placed, 109 completed

Buy & Hold Return: -5.48% (Excess: +2.10%)



BTC DCA -- 3mo Window

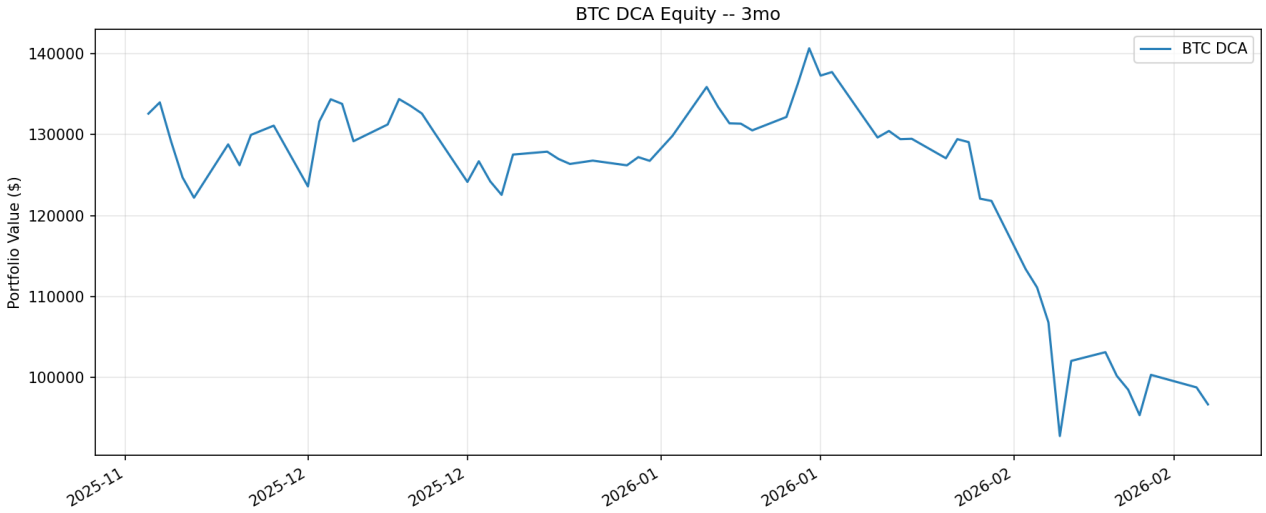
Return: -4.40% (Annualized: -16.46%)

Sharpe Ratio: -2.124

Max Drawdown: 34.05%

Pairs: 57 placed, 55 completed

Buy & Hold Return: -5.48% (Excess: +1.09%)



BTC DCA -- 1mo Window

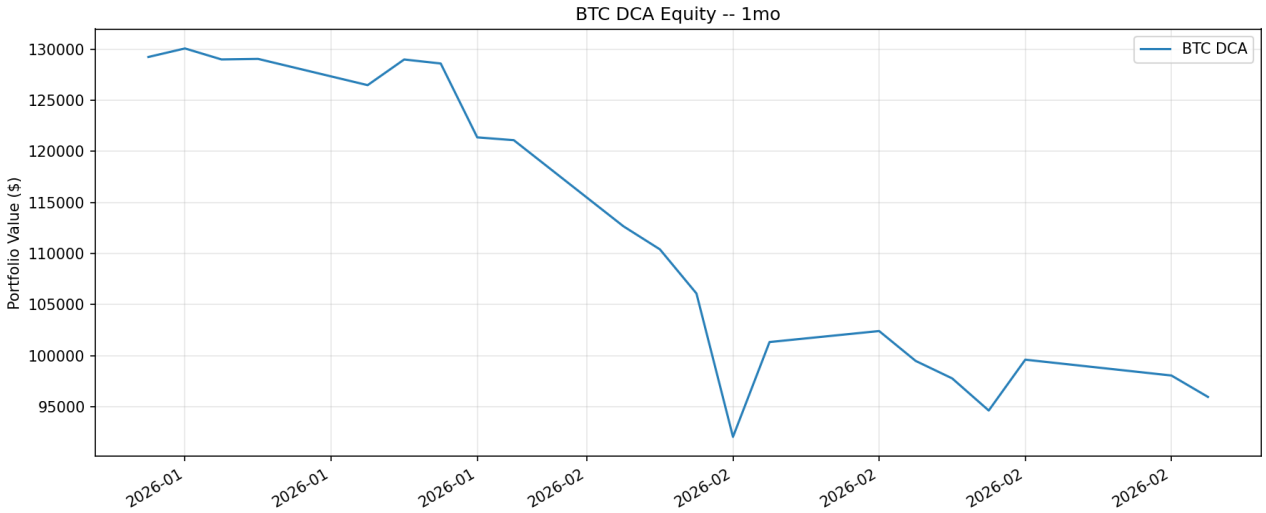
Return: -5.11% (Annualized: -46.70%)

Sharpe Ratio: -4.795

Max Drawdown: 29.24%

Pairs: 20 placed, 19 completed

Buy & Hold Return: -5.48% (Excess: +0.38%)



3. Growth Benchmarks

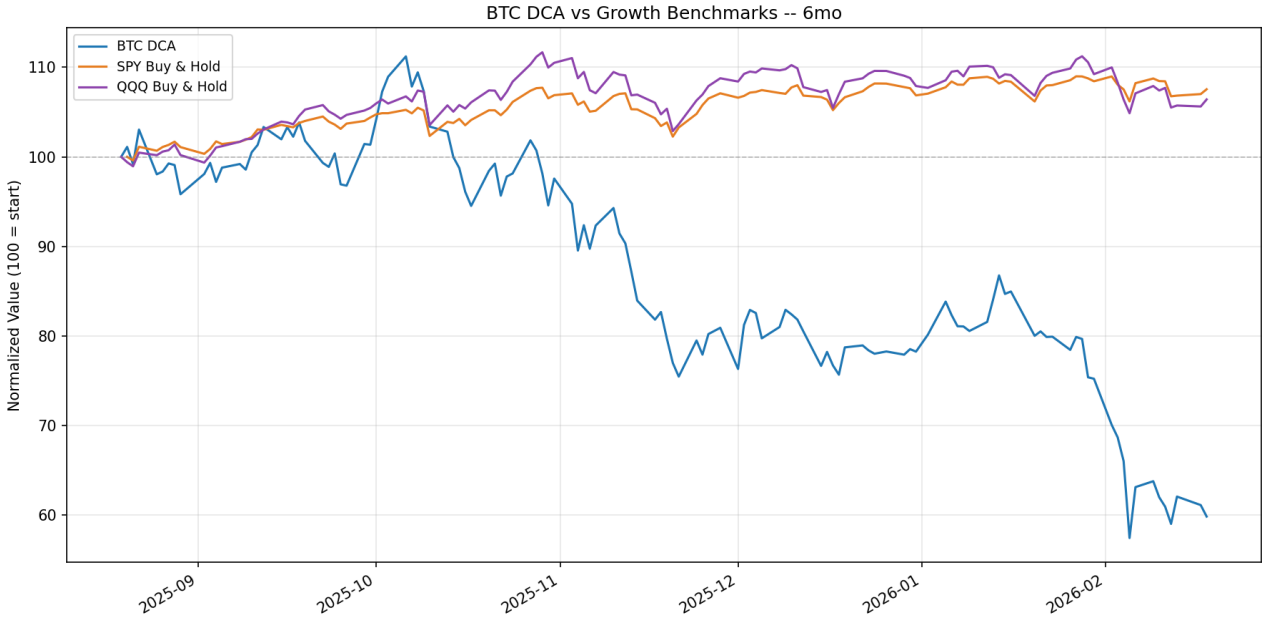
SPY (S&P 500) and QQQ (Nasdaq 100) buy-and-hold returns provide context for BTC DCA performance. All portfolios are normalized to the same initial capital for fair comparison. These benchmarks represent passive equity exposure without any active management or hedging.

Benchmarks -- 6mo Window

BTC DCA: -3.38%, Sharpe -2.028

SPY Buy & Hold: +7.55%, Sharpe 0.904, MaxDD 5.07%

QQQ Buy & Hold: +6.41%, Sharpe 0.555, MaxDD 7.88%

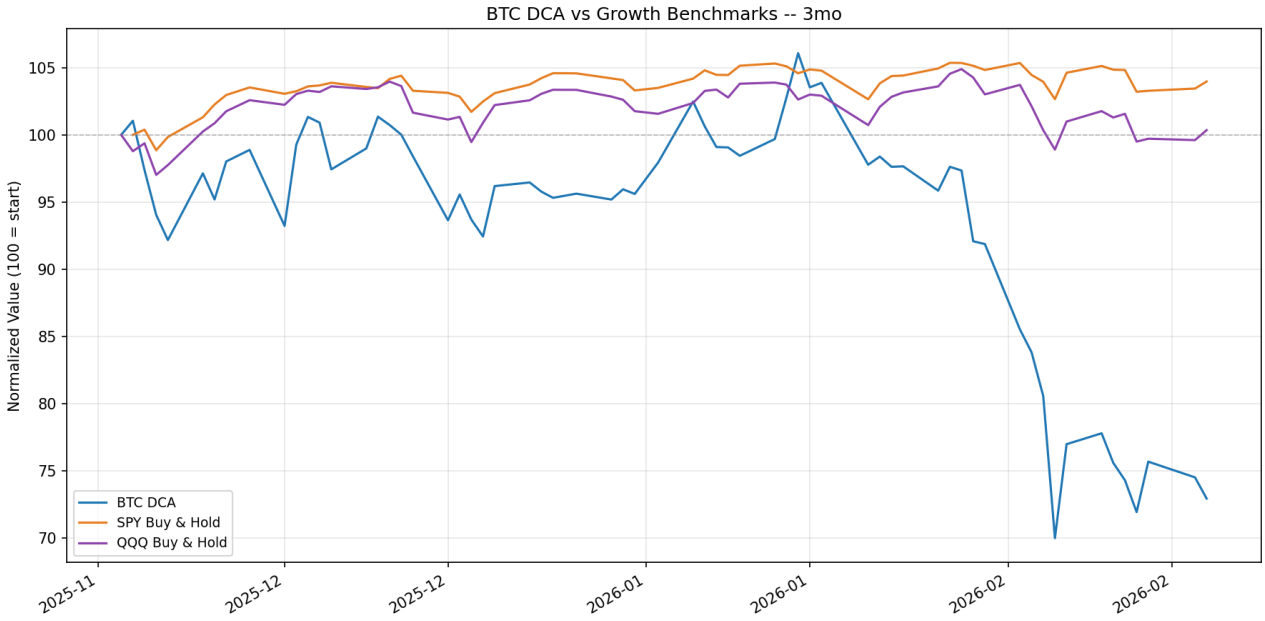


Benchmarks -- 3mo Window

BTC DCA: -4.40%, Sharpe -2.124

SPY Buy & Hold: +3.97%, Sharpe 1.023, MaxDD 2.58%

QQQ Buy & Hold: +0.35%, Sharpe -0.143, MaxDD 5.72%

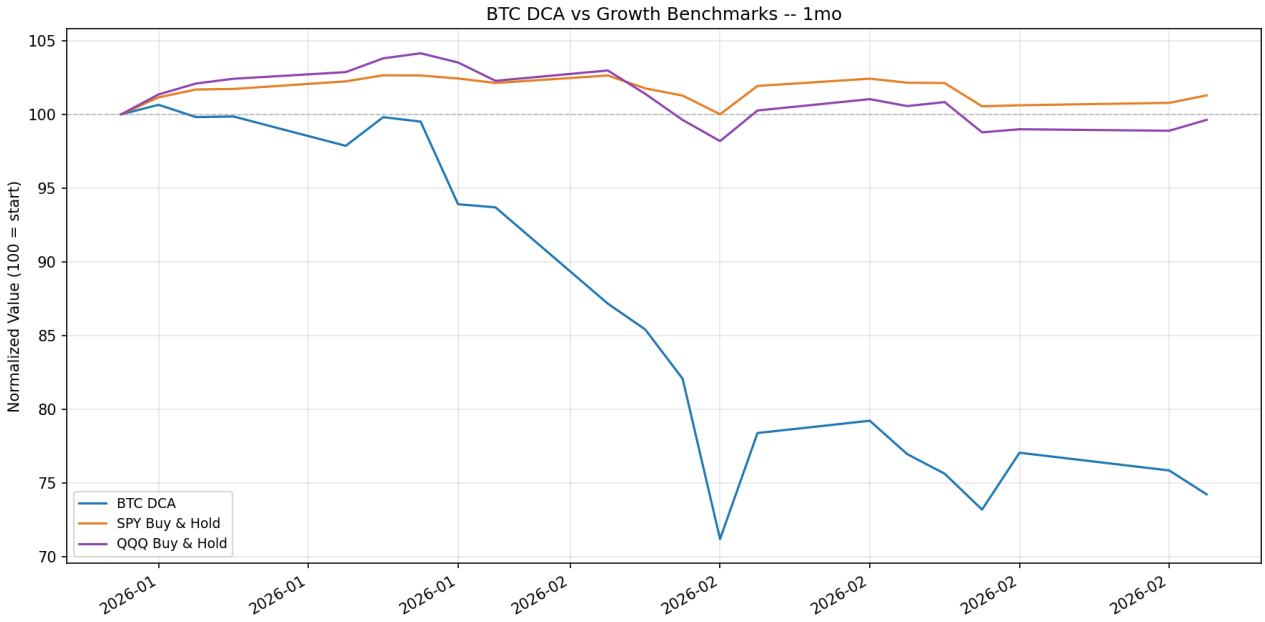


Benchmarks -- 1mo Window

BTC DCA: -5.11%, Sharpe -4.795

SPY Buy & Hold: +1.29%, Sharpe 0.958, MaxDD 2.57%

QQQ Buy & Hold: -0.37%, Sharpe -0.467, MaxDD 5.72%



4. Drawdown Strategies

Three strategies designed for continued BTC drawdown scenarios. Each modifies the base DCA approach to limit losses during sustained declines.

Cash Reserve

Pauses new DCA pair placement when the portfolio drops >10% from its rolling peak. Resumes when price recovers 5% from the trough. Existing orders continue to execute -- only new pair creation is halted.

6mo Window

Return: -41.01%, Sharpe: -2.048, MaxDD: 49.05%

Pause count: 10, Pairs: 41/42

3mo Window

Return: -27.26%, Sharpe: -2.139, MaxDD: 34.12%

Pause count: 3, Pairs: 44/45

1mo Window

Return: -25.89%, Sharpe: -4.825, MaxDD: 29.28%

Pause count: 3, Pairs: 10/10

Trailing Stop

Full position exit via an 8% trailing stop from the rolling price peak. Re-enters when price drops 5% below the exit price. This is a pure position-level strategy -- no DCA pairing.

6mo Window

Return: -25.50%, Sharpe: -1.602, MaxDD: 34.53%

Exits: 6, Re-entries: 5

3mo Window

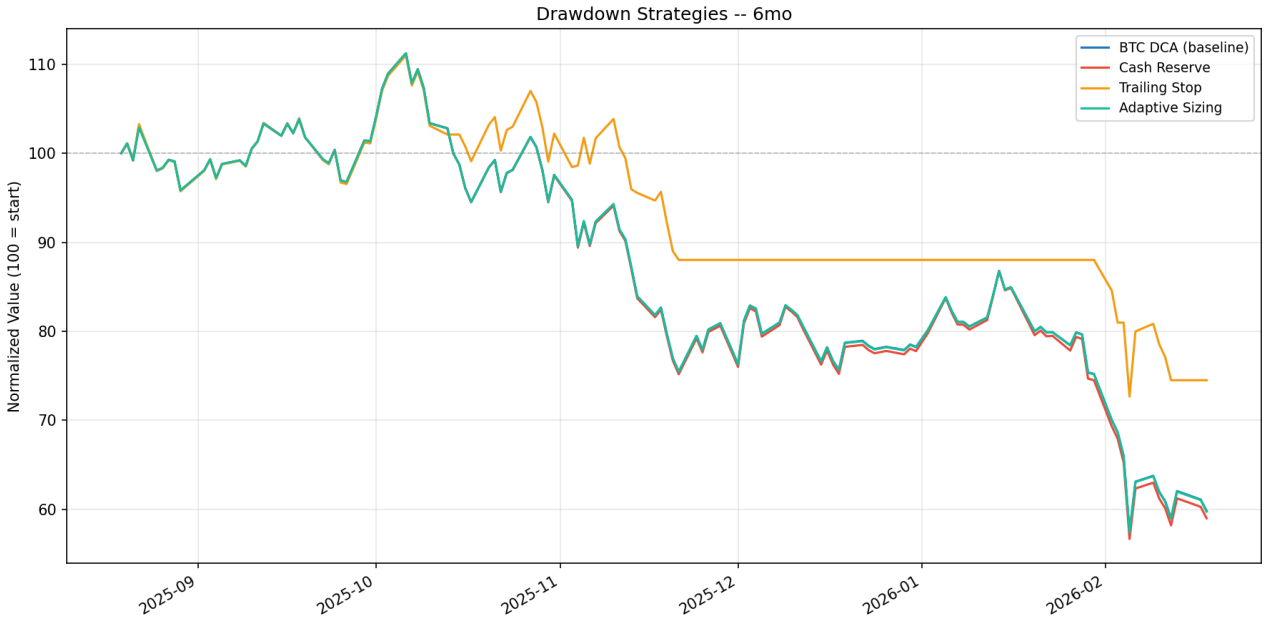
Return: -21.32%, Sharpe: -2.817, MaxDD: 24.05%

Exits: 3, Re-entries: 2

1mo Window

Return: -19.41%, Sharpe: -4.637, MaxDD: 21.77%

Exits: 3, Re-entries: 2



Adaptive Sizing

Runs the DCA strategy but dynamically scales lot size inversely with rolling 21-day volatility. High vol = smaller lots (less whipsaw exposure), low vol = larger lots (capture more of the move).

6mo Window

Return: -40.26%, Sharpe: -2.033, MaxDD: 48.41%

Avg lot size: 91, Median vol: 42.73%

3mo Window

Return: -27.09%, Sharpe: -2.126, MaxDD: 33.99%

Avg lot size: 103, Median vol: 50.12%

1mo Window

Return: -25.72%, Sharpe: -4.797, MaxDD: 29.15%

Avg lot size: 110, Median vol: 47.10%

Drawdown Strategies -- 6mo
BTC DCA (baseline)
Cash Reserve
Trailing Stop
Adaptive Sizing
Normalized Value (100 = start)
110
100
90
80
70
60
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02

5. BTC DCA + SPY Protective Put

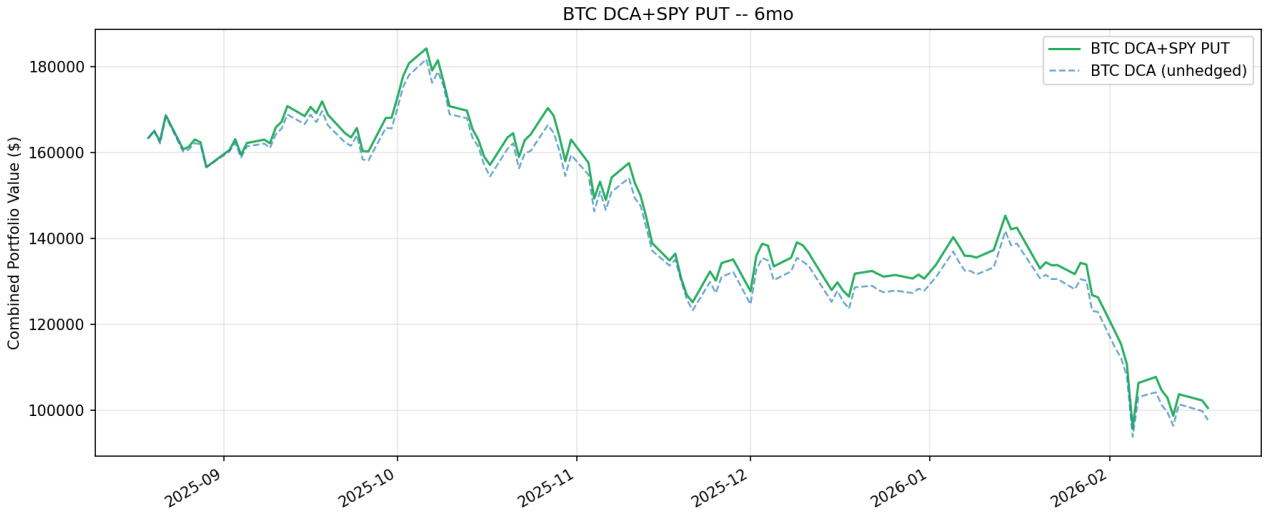
This section shows BTC DCA portfolio combined with SPY protective puts. The put hedge provides downside protection on the broad market, partially offsetting BTC declines correlated with equity sell-offs.

6mo Window

Combined Return: -38.46%

Sharpe Ratio: -1.857

Max Drawdown: 47.97%

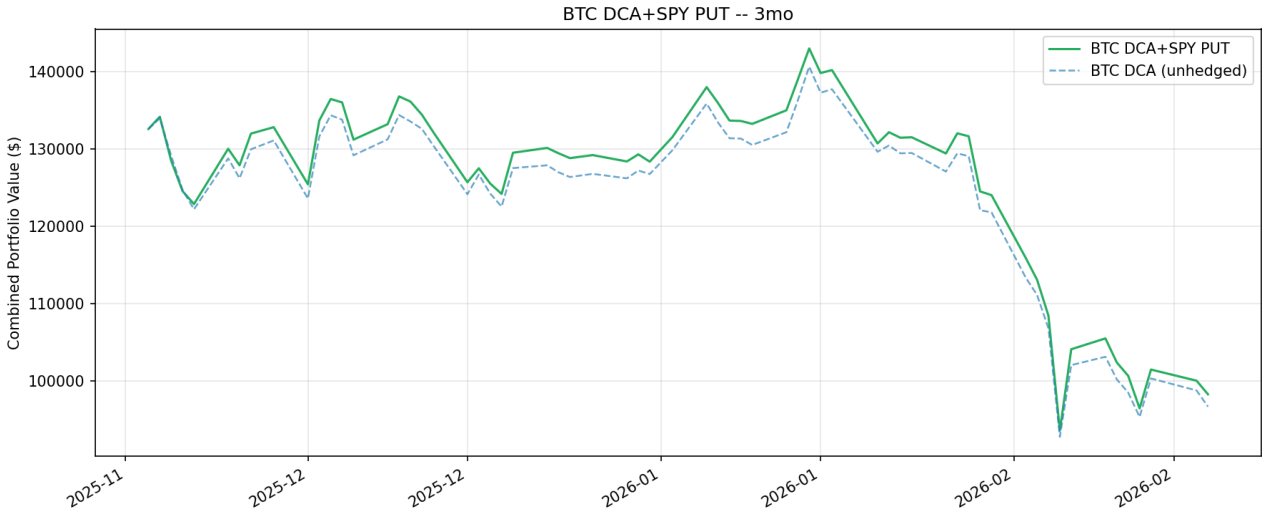


3mo Window

Combined Return: -25.88%

Sharpe Ratio: -1.900

Max Drawdown: 34.57%

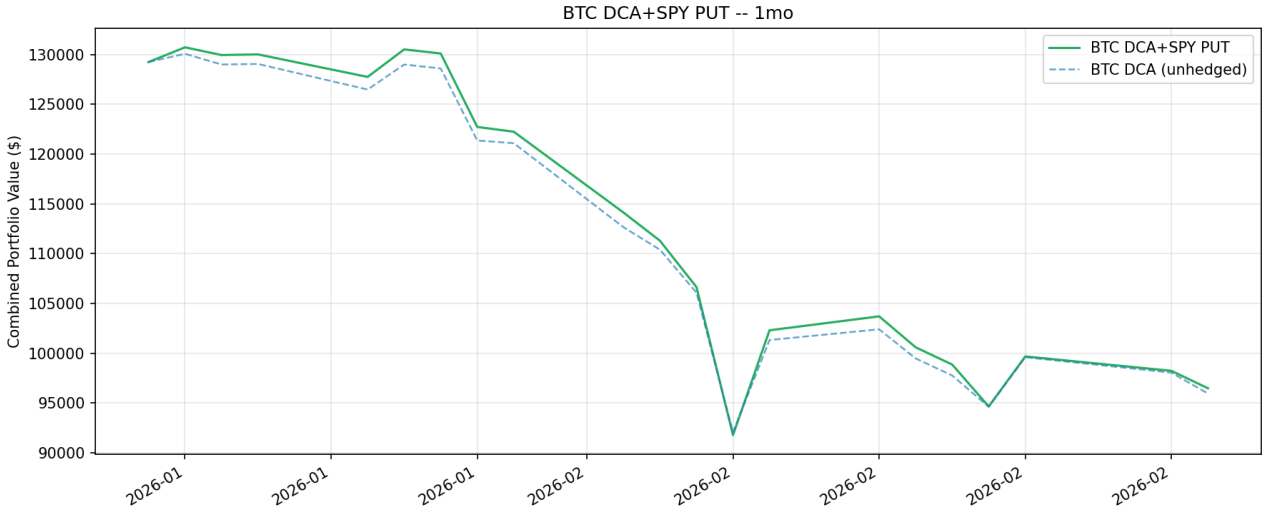


1mo Window

Combined Return: -25.34%

Sharpe Ratio: -4.342

Max Drawdown: 29.78%



6. BTC DCA + IWM Protective Put

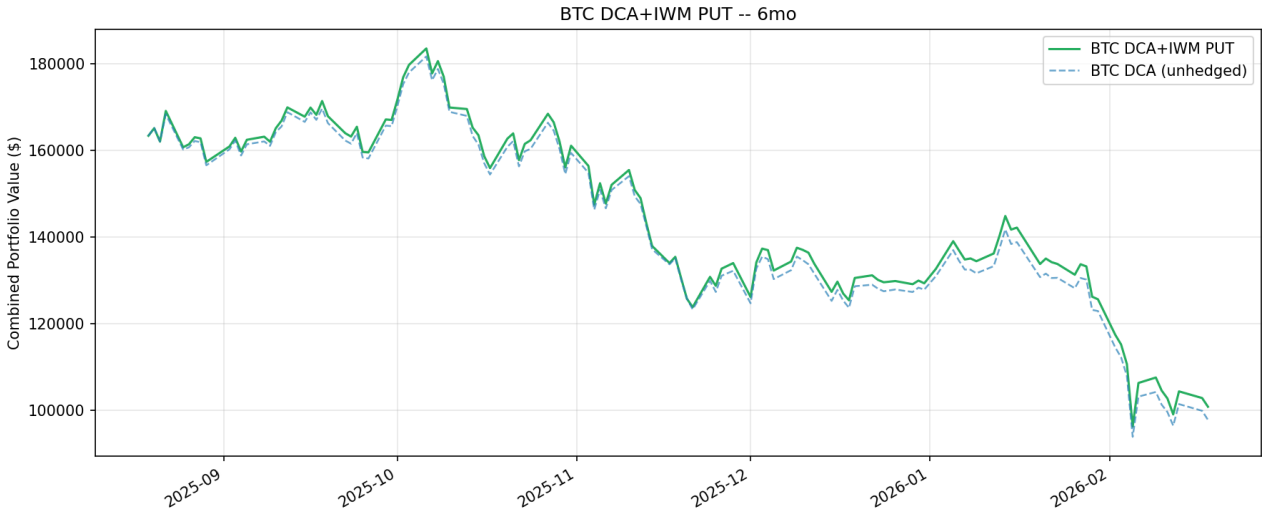
This section shows BTC DCA portfolio combined with IWM protective puts. IWM (Russell 2000) tends to be more volatile than SPY, potentially providing stronger hedging signals during risk-off episodes.

6mo Window

Combined Return: -38.32%

Sharpe Ratio: -1.850

Max Drawdown: 47.60%

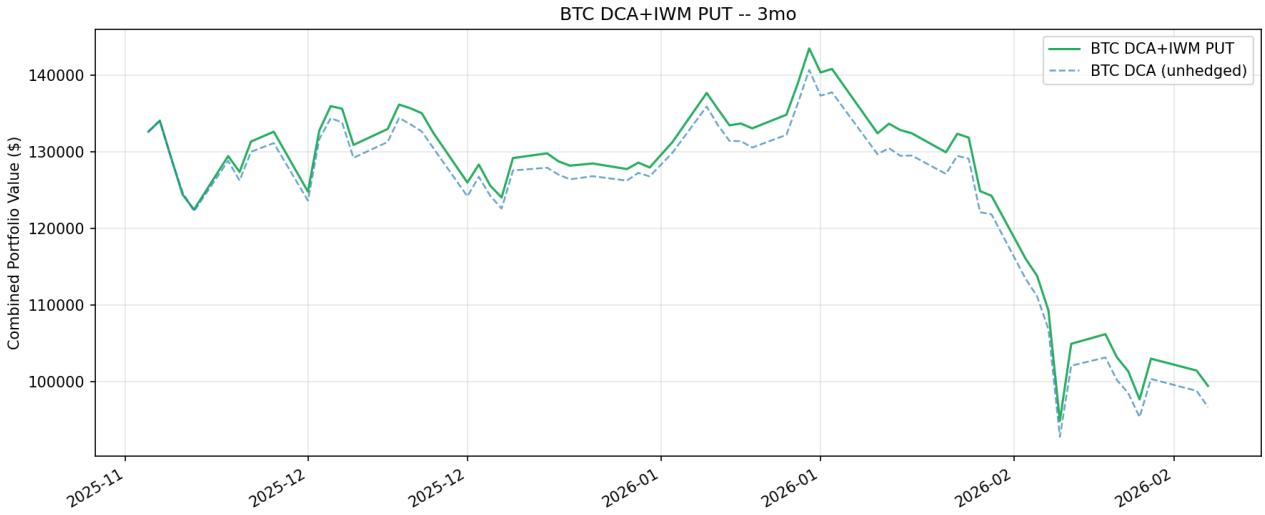


3mo Window

Combined Return: -25.02%

Sharpe Ratio: -1.875

Max Drawdown: 33.89%

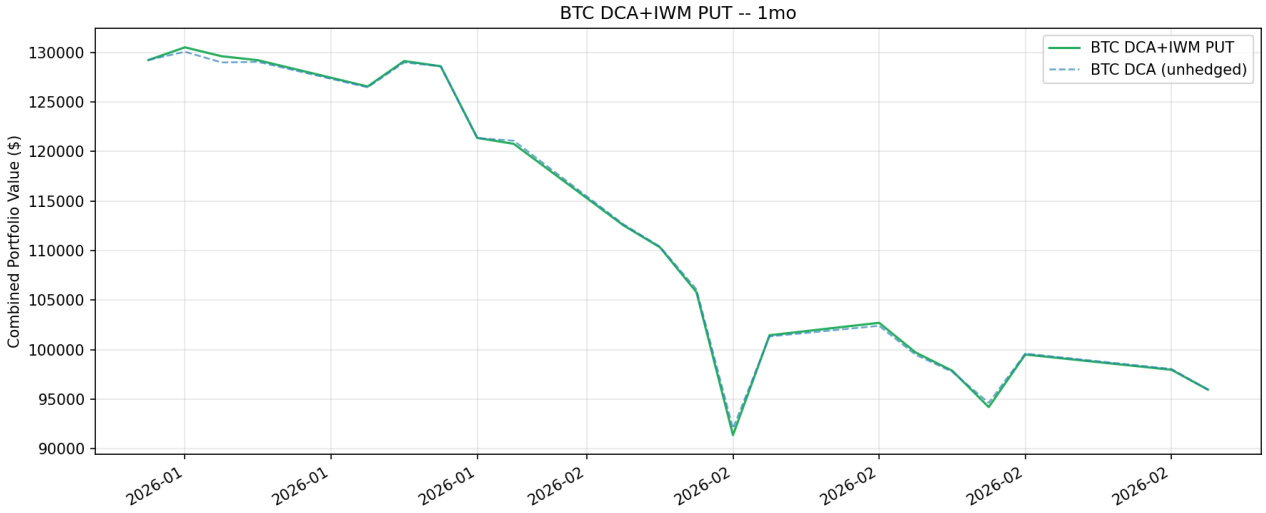


1mo Window

Combined Return: -25.75%

Sharpe Ratio: -4.540

Max Drawdown: 29.99%



7. BTC DCA + SPY Collar

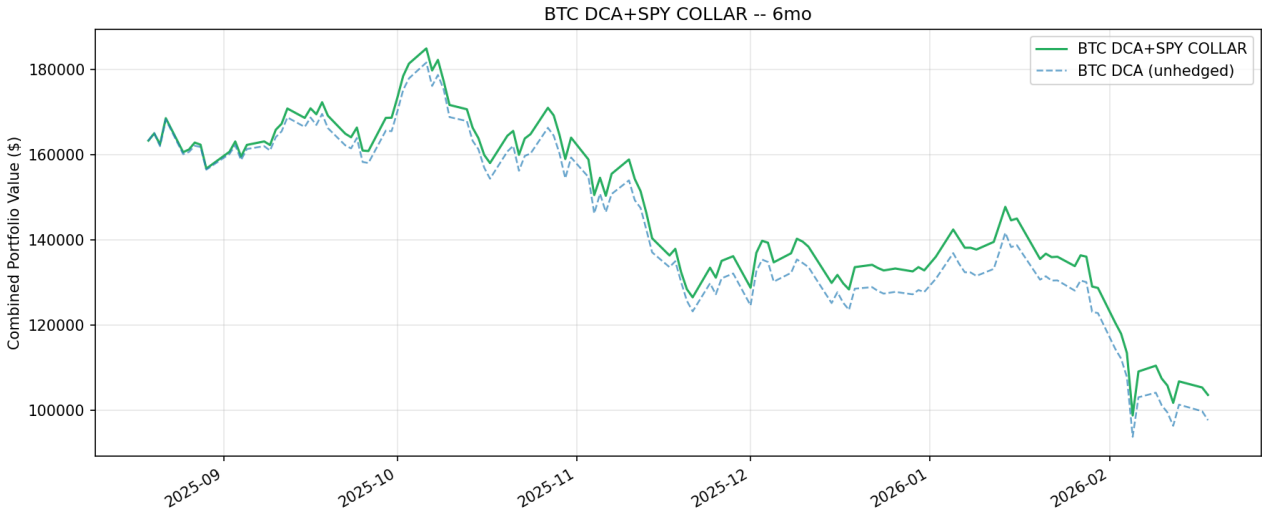
This section shows BTC DCA portfolio combined with SPY collars (long put + short call). The collar reduces net hedging cost by selling upside via the call, but caps the combined portfolio's gains.

6mo Window

Combined Return: -36.55%

Sharpe Ratio: -1.786

Max Drawdown: 46.58%

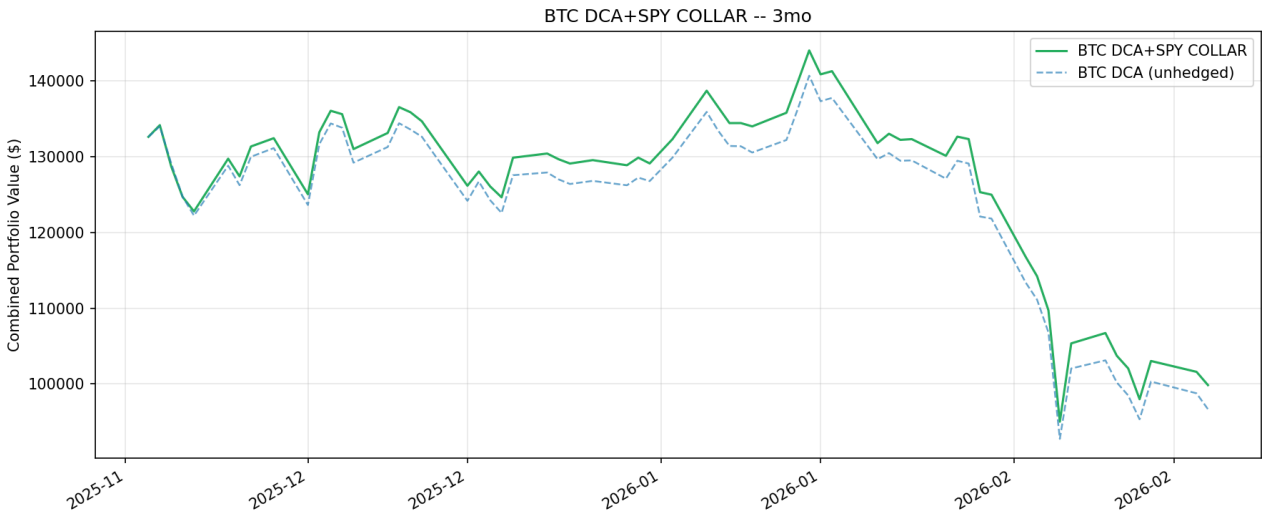


3mo Window

Combined Return: -24.69%

Sharpe Ratio: -1.836

Max Drawdown: 34.00%

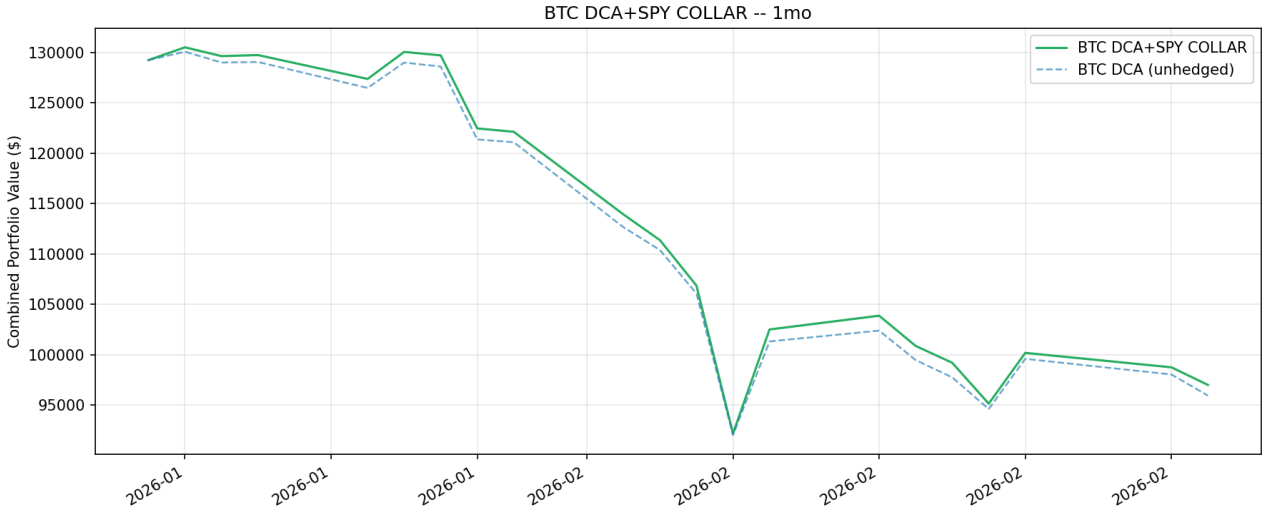


1mo Window

Combined Return: -24.93%

Sharpe Ratio: -4.339

Max Drawdown: 29.36%



8. Standalone Options Strategies

The following results show each options strategy in isolation -- holding the underlying shares with the specified hedge overlay, without any DCA component.

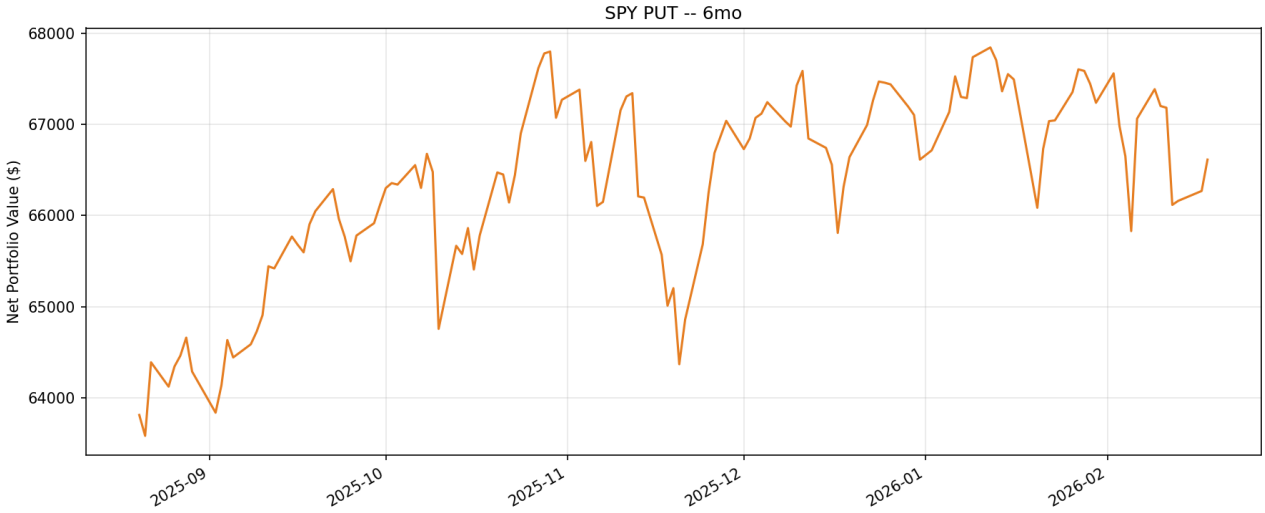
SPY PUT

6mo Window

Return: +4.39% | Sharpe: 0.390 | Max DD: 5.06%

Premium Paid: $2,016.51 | Received: $0.00 | Net Cost: $2,016.51

Put Intrinsic Recovered: $0.00 | Rolls: 6

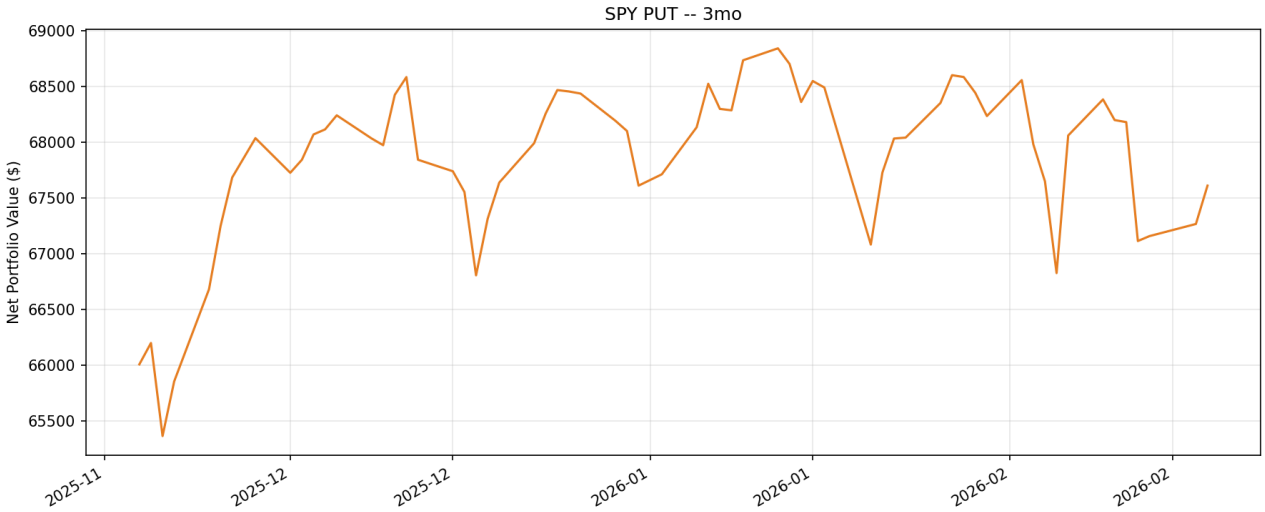


3mo Window

Return: +2.43% | Sharpe: 0.501 | Max DD: 2.93%

Premium Paid: $1,018.74 | Received: $0.00 | Net Cost: $1,018.74

Put Intrinsic Recovered: $0.00 | Rolls: 3

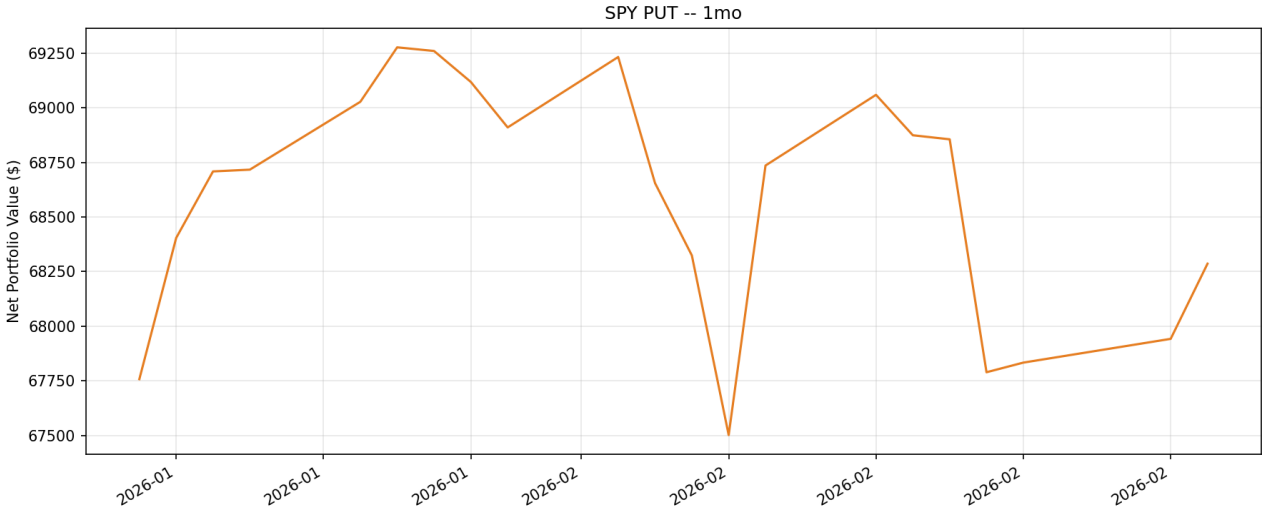


1mo Window

Return: +0.78% | Sharpe: 0.463 | Max DD: 2.56%

Premium Paid: $342.65 | Received: $0.00 | Net Cost: $342.65

Put Intrinsic Recovered: $0.00 | Rolls: 1



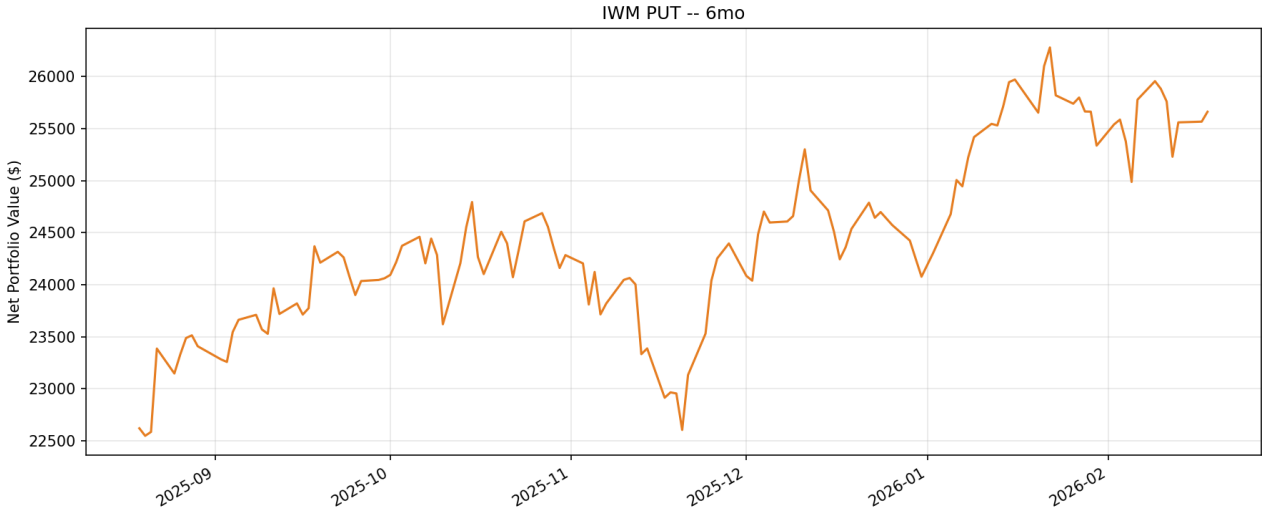
IWM PUT

6mo Window

Return: +13.44% | Sharpe: 1.230 | Max DD: 8.83%

Premium Paid: $736.91 | Received: $0.00 | Net Cost: $736.91

Put Intrinsic Recovered: $0.00 | Rolls: 6

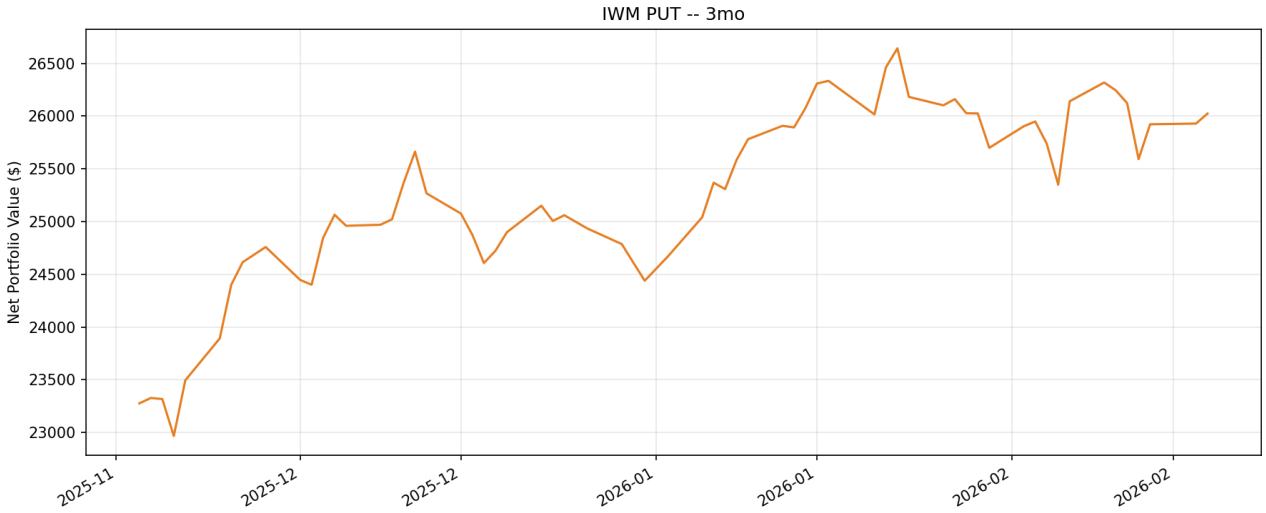


3mo Window

Return: +11.80% | Sharpe: 2.463 | Max DD: 4.85%

Premium Paid: $375.53 | Received: $0.00 | Net Cost: $375.53

Put Intrinsic Recovered: $0.00 | Rolls: 3

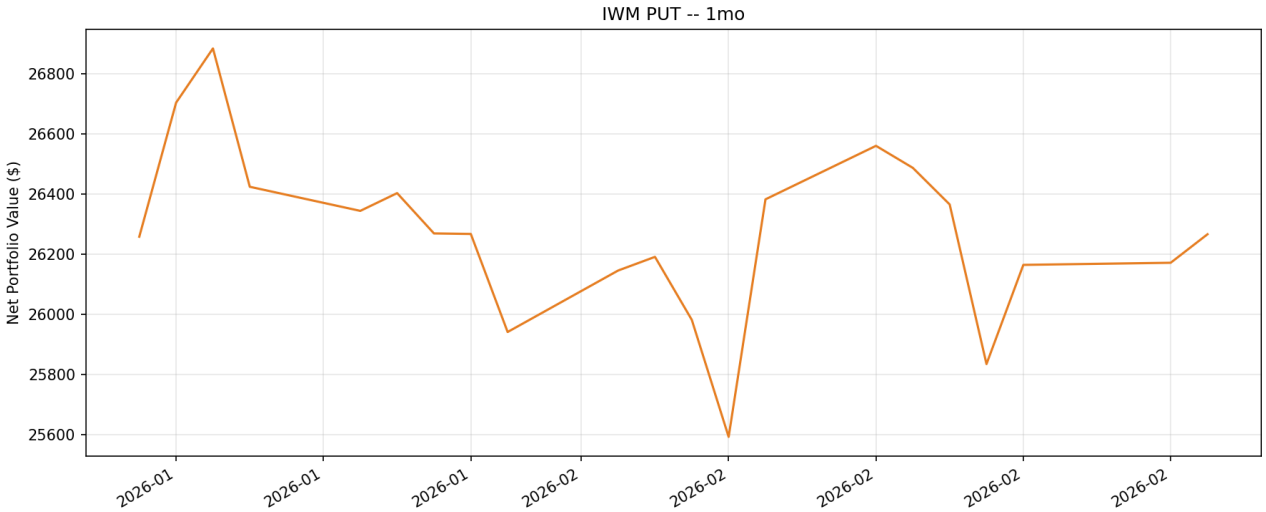


1mo Window

Return: +0.03% | Sharpe: -0.149 | Max DD: 4.81%

Premium Paid: $132.78 | Received: $0.00 | Net Cost: $132.78

Put Intrinsic Recovered: $0.00 | Rolls: 1



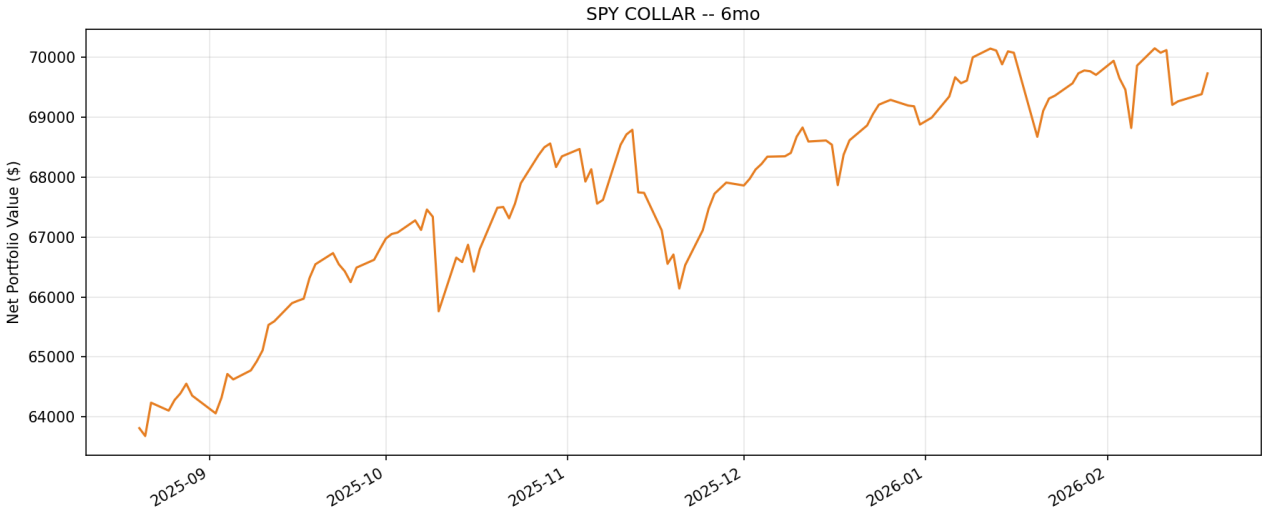
SPY COLLAR

6mo Window

Return: +9.28% | Sharpe: 1.514 | Max DD: 3.85%

Premium Paid: $2,016.51 | Received: $3,121.72 | Net Cost: $-1,105.21

Put Intrinsic Recovered: $0.00 | Rolls: 6

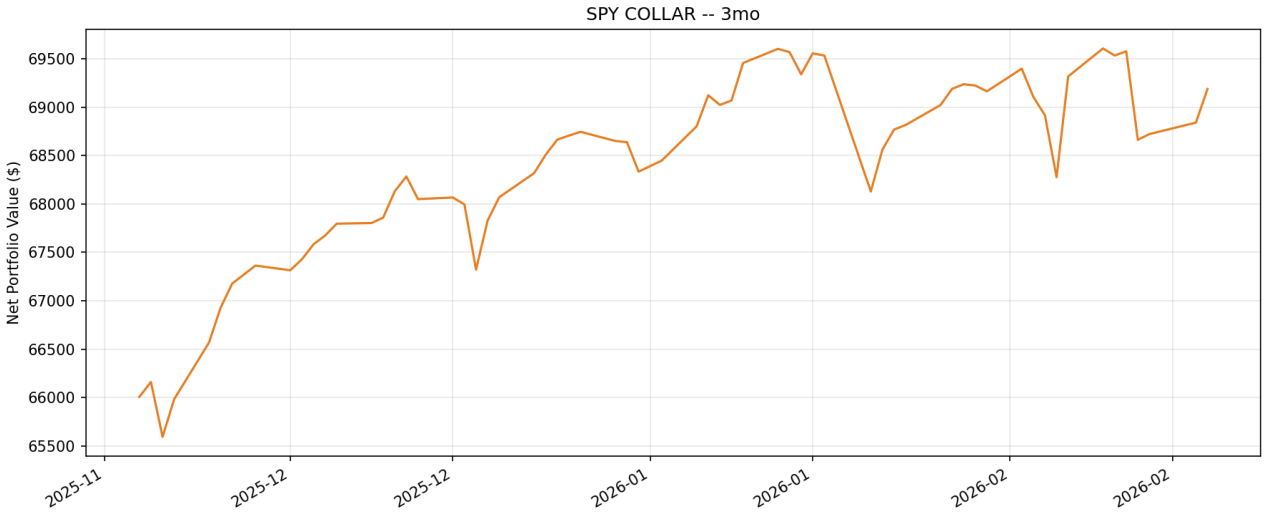


3mo Window

Return: +4.82% | Sharpe: 1.762 | Max DD: 2.12%

Premium Paid: $1,018.74 | Received: $1,577.06 | Net Cost: $-558.32

Put Intrinsic Recovered: $0.00 | Rolls: 3

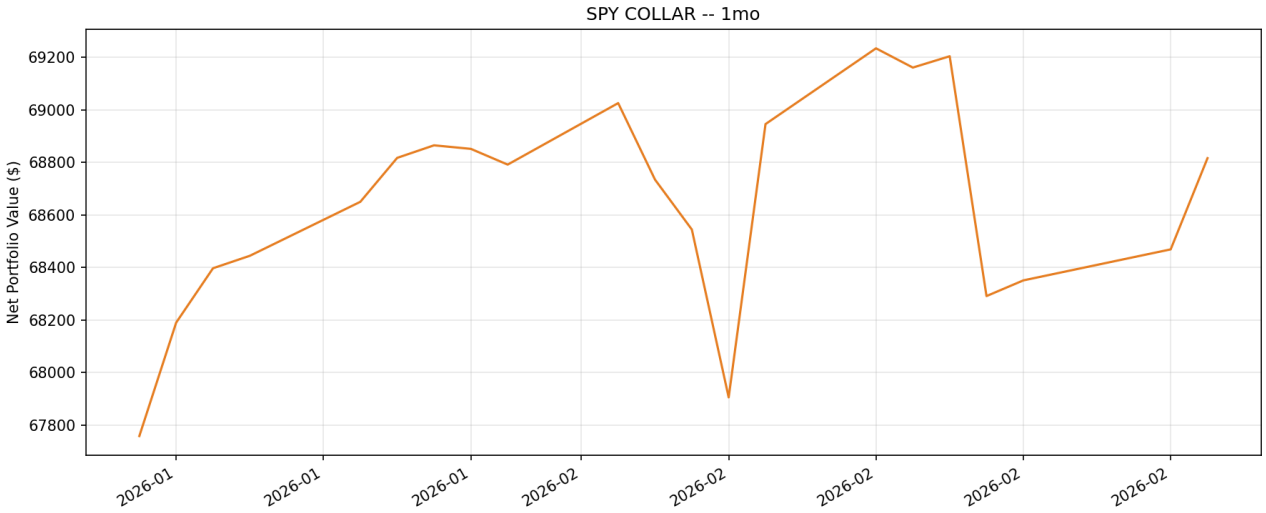


1mo Window

Return: +1.56% | Sharpe: 1.626 | Max DD: 1.62%

Premium Paid: $342.65 | Received: $530.46 | Net Cost: $-187.81

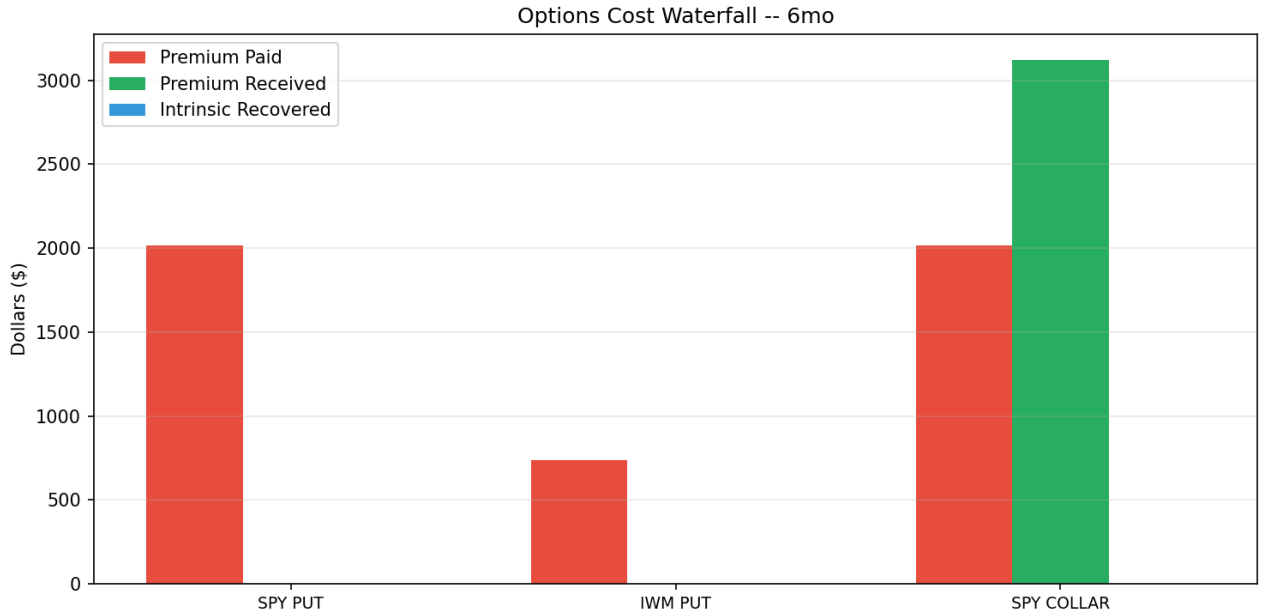
Put Intrinsic Recovered: $0.00 | Rolls: 1

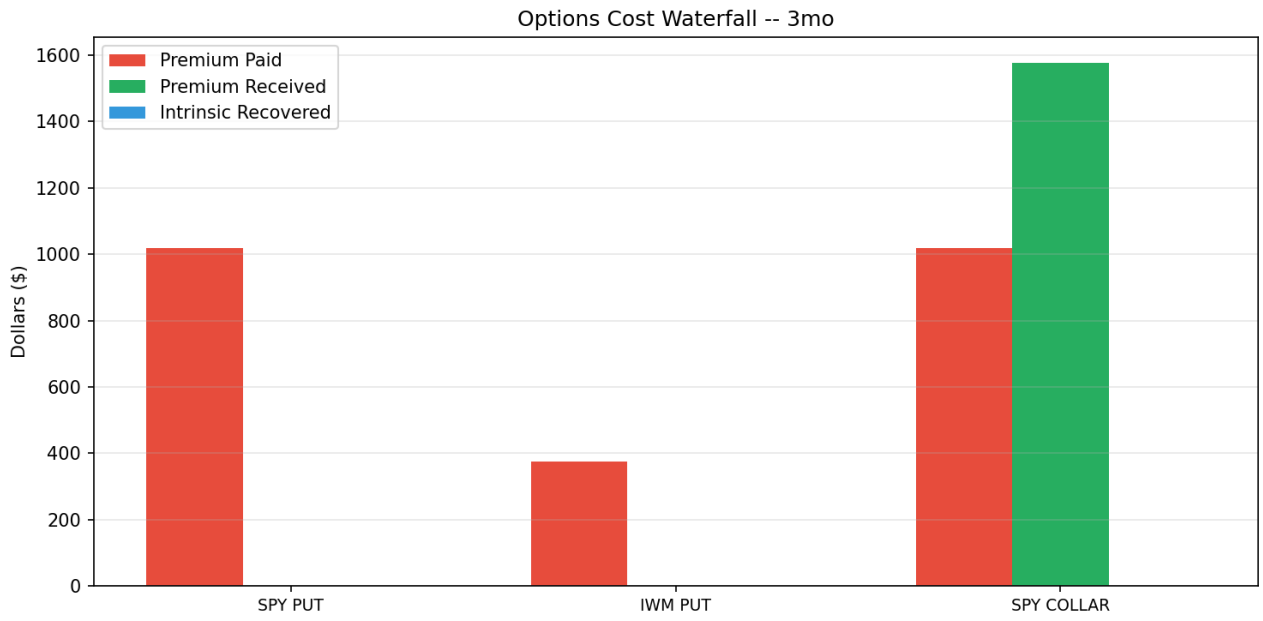

SPY COLLAR -- 1mo
Net Portfolio Value ($)
69200
69000
68800
68600
68400
68200
68000
67800
2026-01
2026-01
2026-01
2026-02
2026-02
2026-02
2026-02
2026-02

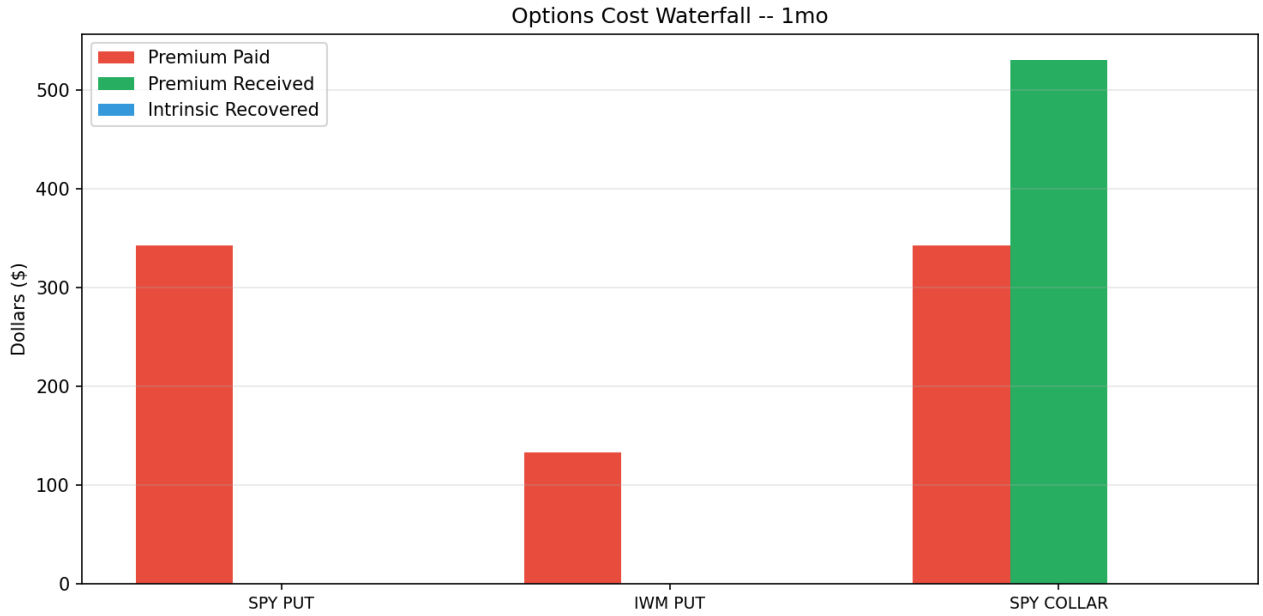
9. Options Cost Analysis

Breakdown of options hedging costs: premiums paid for puts, premiums received from short calls (collar only), intrinsic value recovered at option expiry, and net protection cost as a percentage of initial portfolio value.

Strategy	Window	Prem Paid	Prem Rcvd	Intrinsic	Net Cost	Cost%
SPY PUT	6mo	$2,017	$0	$0	$2,017	3.2%
SPY PUT	3mo	$1,019	$0	$0	$1,019	1.5%
SPY PUT	1mo	$343	$0	$0	$343	0.5%
IWM PUT	6mo	$737	$0	$0	$737	3.3%
IWM PUT	3mo	$376	$0	$0	$376	1.6%
IWM PUT	1mo	$133	$0	$0	$133	0.5%
SPY COLLAR	6mo	$2,017	$3,122	$0	$-1,105	-1.7%
SPY COLLAR	3mo	$1,019	$1,577	$0	$-558	-0.8%
SPY COLLAR	1mo	$343	$530	$0	$-188	-0.3%




Options Cost Waterfall -- 3mo
Premium Paid
Premium Received
Intrinsic Recovered
Dollars ($)
0
200
400
600
800
1000
1200
1400
1600
SPY PUT
IWM PUT
SPY COLLAR


Options Cost Waterfall -- 1mo
Premium Paid
Premium Received
Intrinsic Recovered
Dollars ($)
0
100
200
300
400
500
SPY PUT
IWM PUT
SPY COLLAR

10. Risk Analysis

This section identifies key risks that affect the BTC DCA strategy in real trading but are not captured by the backtest simulation.

Weekend Gap Risk

BTC ETF (NYSE: BTC) only trades Monday-Friday 9:30am-4:00pm ET. Crypto markets trade 24/7. A 10% weekend crypto drop means Monday's open gaps past the stop price -- the sell fills at the gap-down open, which can be significantly worse than the intended stop level. The backtest uses daily bars and does not model this gap risk.

Flash Crash / Intraday Gap Risk

Daily bars compress all intraday price action into a single candle. A flash crash that recovers within the same day may trigger both the sell stop AND the buy limit in the same bar -- overstating pair completion rates. In reality, the crash and recovery may happen too fast for orders to fill at intended prices.

SELL_ONLY Cascade

In sustained drawdowns, sell stops keep triggering but buy limits never fill (price doesn't drop far enough). The position shrinks as shares are sold, and cash accumulates. When the market eventually recovers, you hold fewer shares and miss the bounce. This is the DCA strategy's primary structural weakness -- it can progressively de-risk precisely when you should be adding exposure.

Correlation Breakdown

SPY/IWM hedges assume BTC price movements correlate with equities during stress periods. In crypto-specific sell-offs (regulation, exchange collapse, stablecoin de-peg), correlation with equities drops to near zero and the equity-based hedges provide no protection. The backtest cannot model regime-dependent correlations.

Liquidity and Bid-Ask Spread

BTC ETF spreads widen significantly during high volatility. Real fills will be worse than the simulation assumes (which uses exact stop/limit prices). During extreme moves, market orders may fill several cents away from the intended price, compounding losses across many paired orders.

Tax Basis Complexity

Every completed pair (sell + buy-back) is a taxable event. High completion rates mean a high volume of short-term capital gains/losses to report. Gains are always taxable -- the wash sale rule does NOT defer gains. However, the wash sale rule DOES apply to losses: if you sell at a loss and buy back the same security within 30 days, the loss is disallowed and added to the cost basis of the replacement shares. Since DCA pairs buy back immediately, virtually every losing pair triggers a wash sale, deferring the tax benefit of those losses.

11. Backtest Limitations

Honest assessment of simulation flaws that may cause backtest results to diverge from real trading performance.

Daily bar resolution: Hides intraday dynamics. Stop orders that trigger and recover within a single bar are not modeled accurately.

No weekend/holiday gap modeling: Critical for BTC ETF. Weekend crypto moves cause Monday open gaps that can blow past stop levels.

No slippage or bid-ask spread: All fills occur at exact stop/limit prices. Real fills will be worse, especially during volatile periods.

Same-bar sell+buy triggers: The simulation allows both legs of a pair to fill on the same daily bar, which overstates completion rates.

Fixed IV assumption (20%): Real implied volatility fluctuates between 15-60% for BTC ETF options. Constant IV understates option costs during high-vol periods and overstates them during calm markets.

No transaction costs: Commissions, exchange fees, and regulatory fees are not included. While small per-trade, they accumulate across many pairs.

Limited data history: BTC ETF has ~389 trading days of history (~1.5 years). This is insufficient for statistically robust conclusions about strategy performance across multiple market cycles.

Survival bias: We are only testing assets that still exist and trade today. Failed ETFs, delisted securities, and collapsed exchanges are not included.

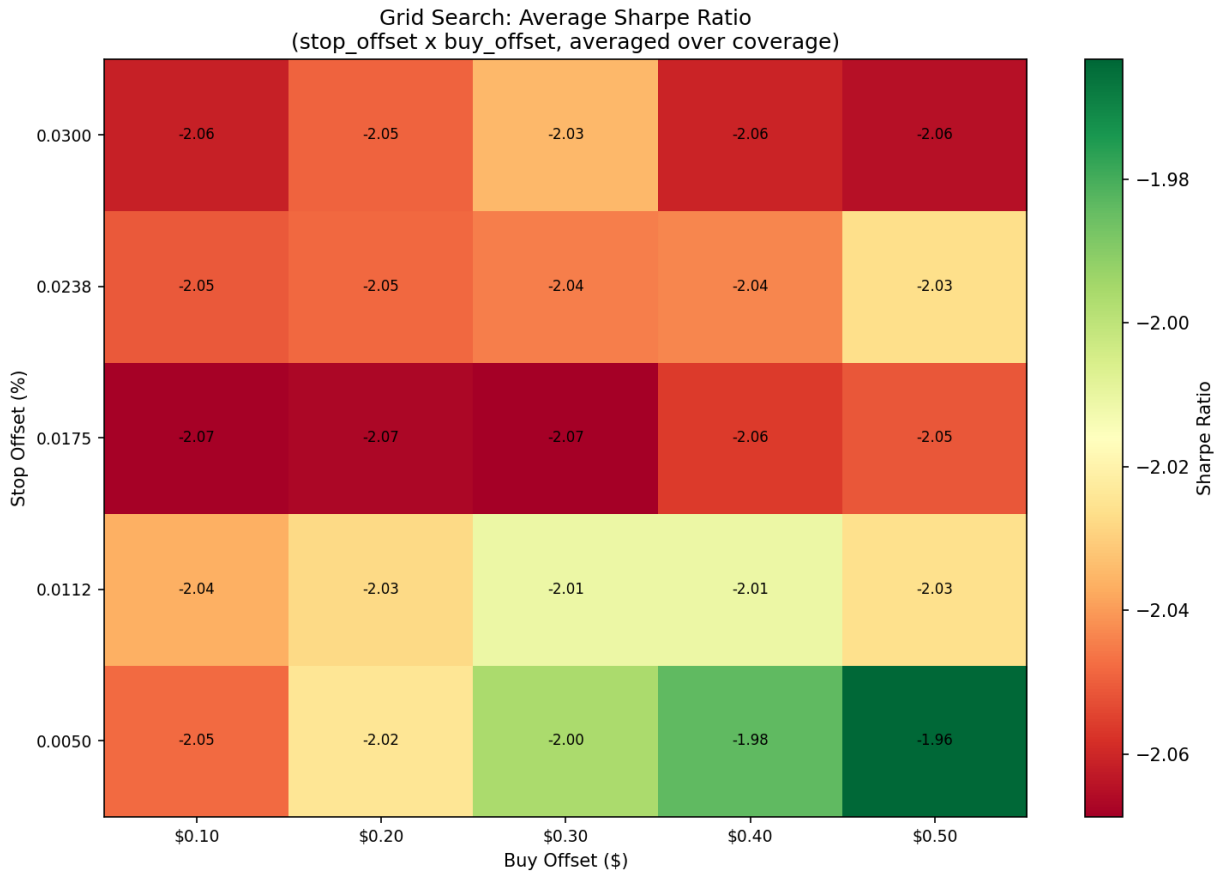
No overnight gap risk: The simulation does not model the risk of overnight gaps in the underlying security that could affect pending orders.

12. Parameter Optimization

Grid search over stop_offset_pct x buy_offset x coverage_threshold to identify parameter combinations that maximize the Sharpe ratio. Total combinations tested: 75.

Top 5 Parameter Combinations by Sharpe

Stop%	Buy$	Cov%	Sharpe	Return%	MaxDD%	Comp%
0.0050	$0.50	30%	-1.951	+0.3	46.5	98
0.0050	$0.50	20%	-1.954	+0.2	46.5	98
0.0050	$0.40	30%	-1.974	-0.9	47.0	98
0.0050	$0.40	20%	-1.976	-0.9	47.0	98
0.0050	$0.50	10%	-1.985	-1.0	47.1	98



Regime-Based Parameter Suggestions

Current regime: HIGH_VOL (rolling vol = 71.7%)

Suggested stop_offset: 0.0200

Suggested buy_offset: $0.35

Suggested coverage: 15%

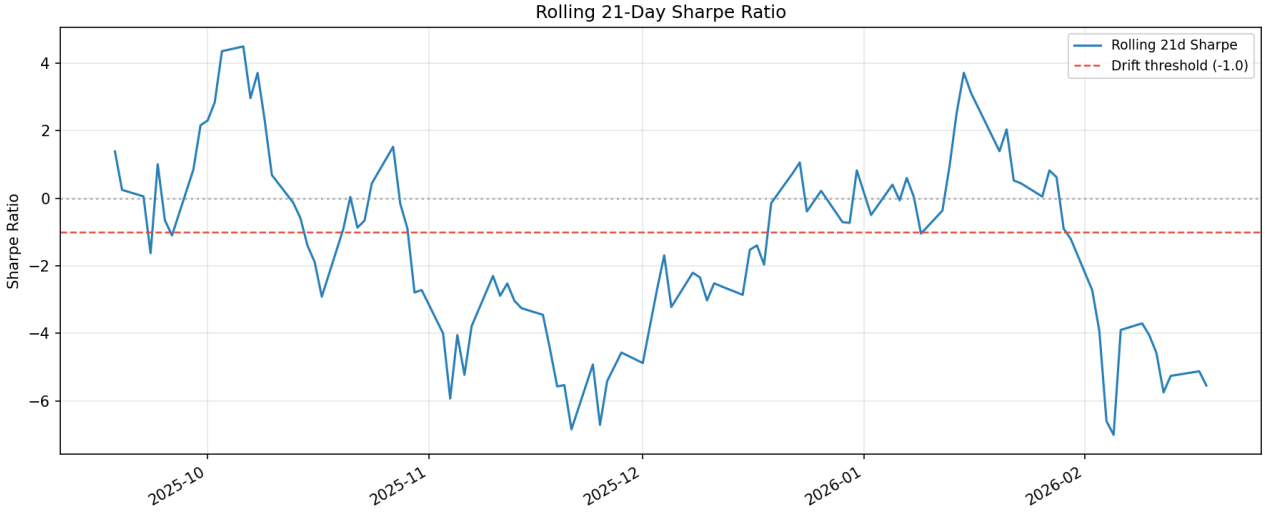
Rationale: Widen stops and buy offsets to avoid whipsaw in high volatility

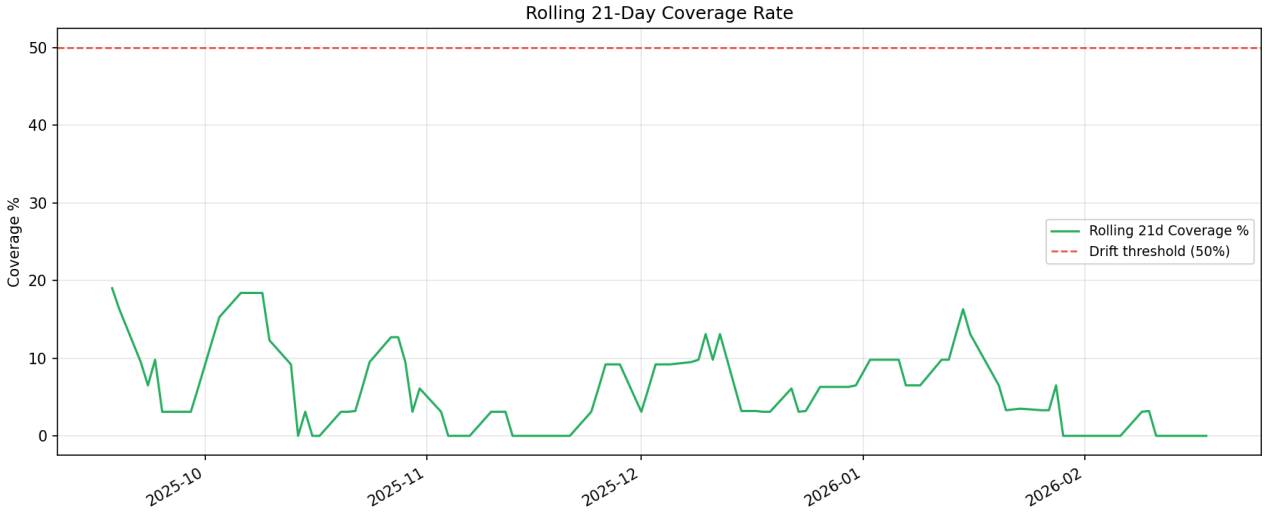
Regime Rules Summary

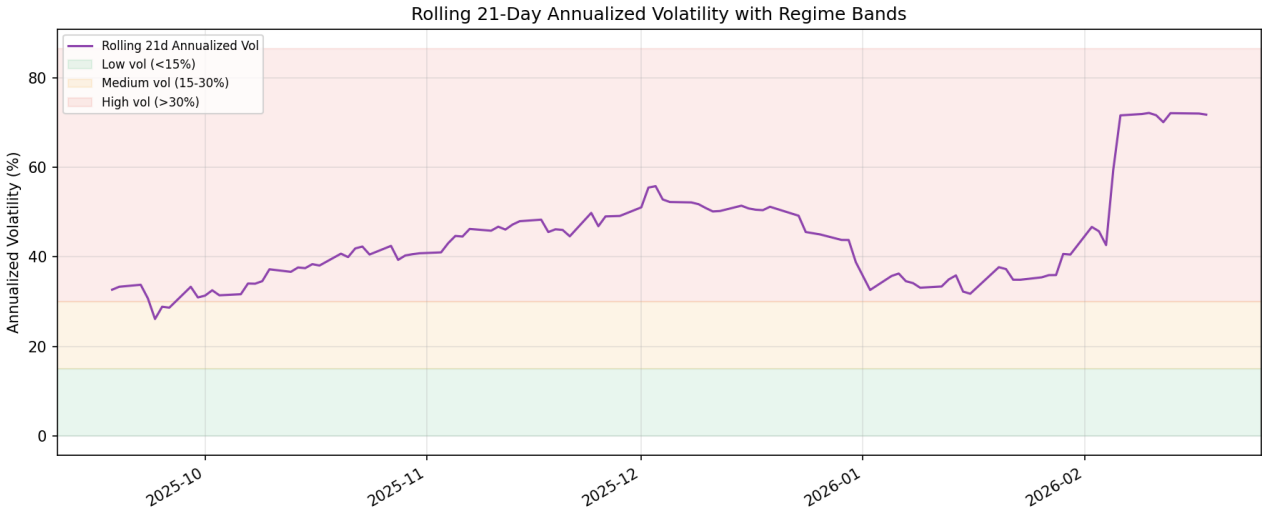
Regime	Stop%	Buy$	Cov%	Condition
LOW_VOL	0.0075	$0.15	25%	Ann. vol < 15%
MEDIUM_VOL	0.0100	$0.20	20%	15% <= vol <= 30%
HIGH_VOL	0.0200	$0.35	15%	Ann. vol > 30%

13. Parameter Monitoring & Drift Detection

Rolling 21-day metrics track strategy health over time. Drift alerts flag dates when metrics cross warning thresholds, indicating the strategy may be underperforming and parameters should be reviewed.







Drift Alerts (159 total)

Date	Metric	Value	Threshold
2025-09-18	rolling completion rate	19.00	50.00
2025-09-19	rolling completion rate	16.30	50.00
2025-09-22	rolling completion rate	9.50	50.00
2025-09-23	rolling sharpe	-1.62	-1.00
2025-09-23	rolling completion rate	6.50	50.00
2025-09-24	rolling completion rate	9.80	50.00
2025-09-25	rolling completion rate	3.10	50.00
2025-09-26	rolling sharpe	-1.10	-1.00
2025-09-26	rolling completion rate	3.10	50.00
2025-09-29	rolling completion rate	3.10	50.00
2025-09-30	rolling completion rate	6.10	50.00
2025-10-01	rolling completion rate	9.20	50.00
2025-10-02	rolling completion rate	12.30	50.00
2025-10-03	rolling completion rate	15.30	50.00
2025-10-06	rolling completion rate	18.40	50.00
2025-10-07	rolling completion rate	18.40	50.00
2025-10-08	rolling completion rate	18.40	50.00
2025-10-09	rolling completion rate	18.40	50.00
2025-10-10	rolling completion rate	12.30	50.00
2025-10-13	rolling completion rate	9.20	50.00

... and 139 more alerts (truncated).

Recommended Monitoring Rules

Review parameters when rolling 21-day Sharpe drops below -1.0 for 3+ consecutive days

Investigate if rolling coverage rate falls below 50% -- may indicate SELL_ONLY cascade

Switch to HIGH_VOL regime parameters when annualized volatility exceeds 30%

Switch to LOW_VOL regime parameters when annualized volatility drops below 15%

Run grid search monthly on latest 126-day window to check for parameter drift

Consider pausing the strategy entirely if max drawdown exceeds 15% in any 21-day window

14. Methodology

Black-Scholes Assumptions

European-style options pricing (Black-Scholes model)

Constant implied volatility (default 20% annualized)

Risk-free rate: 5% annualized

No transaction costs or bid-ask spread on options

Monthly rolling: options expire every 21 trading days and are re-struck

Strike Selection

Put strike: underlying close * (1 - OTM%), default 5% OTM

Call strike (collar): underlying close * (1 + OTM%), default 5% OTM

Strikes re-set at each roll to current market price

Portfolio Construction

BTC DCA: Pairwise sell-stop / buy-limit strategy on BTC ETF shares

Options hedge: 100 shares of SPY or IWM with monthly put or collar overlay

Combined portfolio: BTC DCA net_value + options strategy net P&L

The hedge is funded from a separate capital allocation

No delta hedging or dynamic adjustment between roll dates

Drawdown Strategies

Cash Reserve: Pauses DCA pairing when portfolio drops >10% from peak; resumes on 5% recovery

Trailing Stop: Full exit at 8% trailing stop from peak; re-entry at 5% below exit price

Adaptive Sizing: Scales lot size inversely with 21-day rolling volatility (0.25x-2.0x range)

Parameter Optimization

Grid search: 5 stop_offsets x 5 buy_offsets x 3 coverages = 75 parameter combos

Objective: maximize annualized Sharpe ratio

Rolling metrics: 21-day rolling window for Sharpe, completion rate, and volatility

Drift detection: alerts when rolling Sharpe < -1.0 or coverage < 50%

Regime suggestions: parameter adjustments based on volatility bands (low/medium/high)

Normal CDF Approximation

To avoid a scipy dependency, the Black-Scholes implementation uses the Abramowitz & Stegun rational approximation (formula 7.1.26) for the standard normal CDF, with maximum error of approximately 1.5e-7.

15. Conclusion

This analysis compared BTC DCA performance with and without options-based hedging overlays, drawdown-protective strategies, and growth benchmarks across multiple time windows. Key findings:

6mo: BTC DCA returned -3.4%

vs SPY: +7.5% (BTC DCA -10.9% relative)

vs QQQ: +6.4% (BTC DCA -9.8% relative)

BTC DCA+SPY PUT: -38.5% (-35.1% vs unhedged)

BTC DCA+IWM PUT: -38.3% (-34.9% vs unhedged)

BTC DCA+SPY COLLAR: -36.5% (-33.2% vs unhedged)

3mo: BTC DCA returned -4.4%

vs SPY: +4.0% (BTC DCA -8.4% relative)

vs QQQ: +0.3% (BTC DCA -4.8% relative)

BTC DCA+SPY PUT: -25.9% (-21.5% vs unhedged)

BTC DCA+IWM PUT: -25.0% (-20.6% vs unhedged)

BTC DCA+SPY COLLAR: -24.7% (-20.3% vs unhedged)

1mo: BTC DCA returned -5.1%

vs SPY: +1.3% (BTC DCA -6.4% relative)

vs QQQ: -0.4% (BTC DCA -4.7% relative)

BTC DCA+SPY PUT: -25.3% (-20.2% vs unhedged)

BTC DCA+IWM PUT: -25.8% (-20.6% vs unhedged)

BTC DCA+SPY COLLAR: -24.9% (-19.8% vs unhedged)

Drawdown strategy results:

Cash Reserve (6mo): -41.0%, Sharpe -2.048

Trailing Stop (6mo): -25.5%, Sharpe -1.602

Adaptive Sizing (6mo): -40.3%, Sharpe -2.033

Parameter optimization identified optimal parameters: stop_offset=0.0050, buy_offset=$0.50, coverage=30% (Sharpe -1.951).

Options hedging provides downside protection but at a cost. Protective puts reduce max drawdown during market declines but create a premium drag during calm markets. Collars offset this cost by selling upside, making them more efficient hedges when the underlying trades sideways.

The drawdown strategies offer alternatives to hedging: the Cash Reserve strategy avoids committing new capital during drawdowns, while the Trailing Stop provides a hard exit mechanism. The Adaptive Sizing strategy dynamically adjusts exposure based on market conditions.

IMPORTANT: These results should be interpreted with caution given the significant backtest limitations documented in Section 11. Weekend gap risk, slippage, and the limited data history all suggest real-world performance will likely be worse than simulated results.